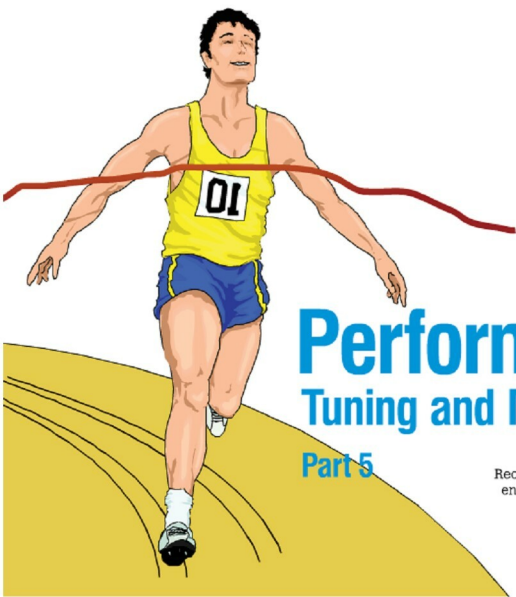




Performance Tuning and Monitoring

Part 5

Reclaim memory, tune swap and enhance network performance.

Last month, we had worked on tuning our memory requirements—changing the kernel's behaviour for memory allocation, exploring the memory overcommit feature, and enhancing performance by tuning the ARP cache. To conclude this series on 'Performance Tuning and Monitoring', we will discuss memory reclamation, tuning swap and enhancing network performance.

States of virtual memory page

The operating system manages the virtual memory pages by reading a label

known as a state. These labels (or states) help the virtual memory to take the required action on the page. So let's first understand the different states of a virtual memory page:

- *FREE* – the page is free or available for immediate allocation.
- *INACTIVE CLEAN* – the page is not active and the contents are being written to the disk. There's nothing to be done on the page. In other words, the page is available for allocation.
- *INACTIVE DIRTY* – 'dirty' in this context means that the contents have been modified after reading from disk but not